

# MAT 1275: College Algebra and Trigonometry (4 Credits)

**Course Description from the Catalog:** An intermediate and advanced algebra course. Topics include quadratic equations, the distance and midpoint formula, graphing parabolas and circles, systems of linear and quadratic equations, an introduction to exponential and logarithmic functions. Topics from trigonometry including basic trigonometric functions, identities, equations and solutions of triangles.

## Course Objectives from the Catalog:

- Simplify and manipulate linear, quadratic, radical, rational, exponential, logarithmic, and trigonometric expressions.
- Solve equations involving linear, quadratic, radical, rational, exponential, logarithmic, or trigonometric expressions as well as systems of linear/quadratic equations.
- Graphically solve equations involving linear and quadratic expressions, including systems of such. Be able to use the unit circle to solve trigonometric equations. Understand the relationships between solutions to equations and their graphs.
- Frame word problems in terms of mathematical equations and/or graphs. Be able to interpret mathematical solutions in terms of the original language of the problem.
- Write solutions of mathematical problems involving linear, quadratic, radical, rational, or trigonometric expressions with full detailed explanations.
- Recognize errors in proposed solutions and explain in written or oral form the nature of such an error as well as be able to correct it.
- Be able to estimate solutions of equations using graphs.

**Optional Textbook:** This can be accessed for free using the following link.

<https://commons.libretexts.org/book/math-93942>

**Companion Workbook:** This can also be accessed for free using the following link.

[https://openlab.citytech.cuny.edu/mat1275co1275/files/2024/01/MAT\\_1275\\_Workbook.pdf](https://openlab.citytech.cuny.edu/mat1275co1275/files/2024/01/MAT_1275_Workbook.pdf)

## Instructor Information

**Instructor Name:** Yao-Rui Yeo

**Instructor Office:** Namm N-604

**Office Hours:** Mondays and Tuesdays, 09:45 – 10:45

**Instructor Email:** YA0-RUI.YE099@CityTech.cuny.edu

## Lecture Information

**Class Semester:** Fall 2026 (Section D109)

**Lecture Location:** Namm N-804

**Lecture Time:** Tuesdays and Thursdays, 08:00 – 09:40

# Course Website

<https://brightspace.cuny.edu/d2l/home/1337167>

## Grading Policy

Your numerical course grade will be calculated using the formula

$$\text{Grade} = 0.1q + 0.2 \max\{m_1, m_2, f\} + 0.2 \max\{m_1, m_3, f\} + 0.2 \max\{m_2, m_3, f\} + 0.3f.$$

Variables are whole numbers between 0 and 100 (inclusive) and explained in the table below.

| Variable | Scored Component            | Date(s)                      |
|----------|-----------------------------|------------------------------|
| $q$      | Quizzes (top 5 scores only) | Most non-midterm Thursdays   |
| $m_1$    | Midterm 1                   | Thursday, September 24, 2026 |
| $m_2$    | Midterm 2                   | Thursday, October 22, 2026   |
| $m_3$    | Midterm 3                   | Thursday, November 19, 2026  |
| $f$      | Final Exam                  | Thursday, December 17, 2026  |

- Grades will be posted on the Course Website. Final letter grade will be assigned according to the CityTech grading scale.
- Quizzes and midterms will test individual work, be in-class, and given on Thursdays. No make-ups will be given. Use the office hours for questions and discussions!
  - Course grade calculation is designed to give you multiple chances for redemption if you perform nonoptimally on, or absolutely have to miss, quizzes or midterms.
- Optional homework problems can be found in the Companion Workbook. It is highly recommended to study them. *Tip: Similar problems may appear in graded work...*
- Electronic devices (e.g. calculators, phones) and external aids (e.g. cheatsheet, ChatGPT) are *forbidden* for any graded component of the course. Using these will result in a score of 0 on the relevant graded component plus additional penalties after a report to the Academic Integrity Office for academic cheating.
- It is important to write down all work in your solution. If you skip essential steps, the instructor reserves the right to take off points.
  - With that said, you are free to use alternative methods instead of those taught in class. In this case, the instructor will grade your solution using an alternative rubric.
- Each quiz contains two problems, each graded 0, 5, or 10 points. Quizzes focus on testing content knowledge over fine precision. Imperfect solutions can still be worth 10 points.
  - 0 point: Solution does not demonstrate understanding of core concepts.
  - 5 points: Solution either incorrectly carries out core concepts or skips essential steps.
  - 10 points: Correct solution modulo small details independent of core concepts.
- Midterms are on a 0-to-100 scale and evaluate both content knowledge and fine precision.
  - During the midterm, you are only allowed to leave the classroom after turning in your paper (and not allowed to re-enter the classroom after).
- Please do not miss the final exam.

## Example Grading

Suppose you are tasked with evaluating the sum

$$\frac{1}{2} + \frac{5}{6}.$$

If this were a quiz problem, here are examples of solutions worth 0, 5, and 10 points.

- 0 point: No evidence of knowing how to add fractions.

$$\frac{1}{2} + \frac{5}{6} = \frac{6}{8}$$

- 5 points: Performed common denominators, but added fractions wrong after.

$$\frac{1}{2} + \frac{5}{6} = \frac{3}{6} + \frac{5}{6} = \frac{8}{12}$$

- 5 points: Correct evaluation but did not show essential steps.

$$\frac{1}{2} + \frac{5}{6} = \frac{4}{3}$$

- 10 points: Correct evaluation – problem did not explicitly ask for lowest terms.

$$\frac{1}{2} + \frac{5}{6} = \frac{3}{6} + \frac{5}{6} = \frac{8}{6}$$

If this were a midterm problem instead, it may be worth 5 points and graded using the following rubric.

$$\begin{aligned} \frac{1}{2} + \frac{5}{6} &= \frac{3}{6} + \frac{5}{6} && (2 \text{ points; common denominator}) \\ &= \frac{8}{6} \left( = \frac{4}{3} \right) && (3 \text{ points; adding fractions with common denominator}) \end{aligned}$$

Note that, unlike quizzes, the midterm problem will almost certainly ask for lowest terms. In this case, 1 of the 3 points in the second step is given for simplifying to lowest terms.

## CityTech Policies

Please refer to the following link for CityTech policies such as academic integrity.

<https://www.citytech.cuny.edu/about-us/policies.aspx>